

SE(3)-LIVOM: Adaptive Multi-LiDAR-Inertial-Visual Odometry and Mapping for the COMFORT Benchmark

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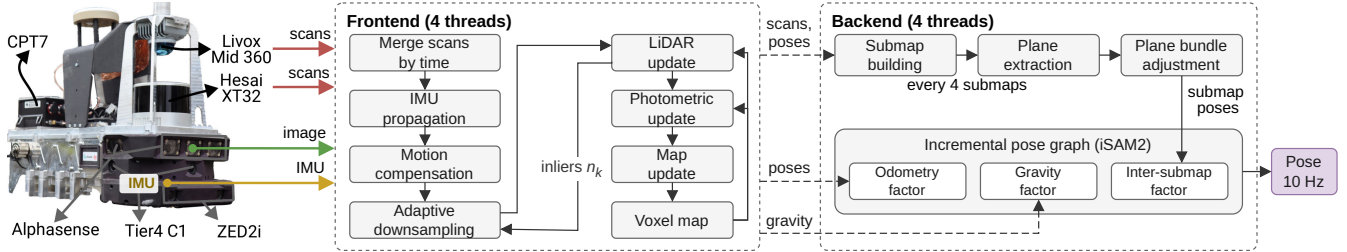


Fig. 1. Pipeline of SE(3)-LIVOM. Red, yellow and green lines carry the LiDAR, IMU and camera data, respectively. The frontend and the backend run in separate threads, and dashed arrows pass data between them. The sensor payload image is adapted from [1].

I. OVERVIEW

SE(3)-LIVOM is our entry to the COMFORT benchmark. It runs in real time on two LiDARs, one camera and an IMU. The system has a frontend and a backend (Fig. 1), and is built on our prior work, SE(3)-LIO [2]. Our contributions are as follows:

- We extend SE(3)-LIO to run with multiple LiDARs and a camera.
- We propose adaptive downsampling to bound computation time within and across scenes.
- We add an online backend with a plane bundle adjustment and an incremental pose graph.

With a single configuration for all scenes, the system reaches an average ATE of 1.40 cm on the six test scenes (Table I). The results can be reproduced with our code at <https://github.com/gunheeShin/se3-livom-comfort>. The information requested by the organizers is in Table III.

II. METHOD

A. Frontend

SE(3)-LIO. The frontend is built on SE(3)-LIO [2]. A LiDAR scan is accumulated while the robot moves. The motion inside the scan is therefore predicted from the IMU first, and the points are then deskewed with that motion. SE(3)-LIO improves both steps. For the prediction, it propagates rotation and translation together on the SE(3) manifold, so the predicted pose stays accurate under aggressive motion. For motion compensation, it treats the predicted motion as uncertain, so the less reliable the pose used to deskew a point, the less that point counts in the update.

Multiple LiDARs. In the corridors of a collapsed building (ARC-2 and ARC-3), every visible surface runs along the motion. With the Hesai LiDAR alone, the frontend loses the

direction along the corridor for up to 19 s at a time, as measured with the localizability of X-ICP [4]. Adding the downward-looking Livox LiDAR reduces this to at most 3 s. The Livox also sees the ground near the robot and lowers the error on the metal stairs of SPX-1. We therefore extend SE(3)-LIO from one LiDAR to two. The points of all sensors are merged into one scan ordered by time, and each point keeps the noise model of its sensor. The two LiDARs see the same surfaces at different times, so an error in the predicted motion now appears as a mismatch between the sensors. The prediction on the SE(3) manifold makes this error smaller, and the point uncertainty reduces the effect of the error that remains. The LiDARs then act as one sensor with a wider field of view.

Photometric update. After the LiDAR update, the camera image refines the same state with the sparse-direct photometric error of FAST-LIVO2 [3]. Image patches are attached to LiDAR map points, and the update aligns the current image to the patches of the points in view.

Adaptive downsampling. A voxel grid with resolution l downsamples each scan before the update. A resolution fine enough for narrow buildings leaves too many points on open terrain, where the update slows down. A resolution coarse enough for open terrain leaves too few matches with the map inside buildings. AdaLIO [5] adapts the resolution to the scene, and X-ICP [4] reads localizability from the correspondences of the registration. We combine the two ideas. At every scan k , the resolution l_k is adjusted so that the number of inliers n_k of the LiDAR update stays near a target N as follows:

$$r_k = \min\left(\max\left((n_k/N)^{1/1.5}, 0.8\right), 1.05\right), \quad (1)$$

$$l_{k+1} = \min(\max(l_k r_k, l_{\min}), l_{\max})$$

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with $N = 2000$, $l_{\min} = 0.1$ m and $l_{\max} = 0.5$ m.

TABLE I. ATE RMSE (cm) on the validation and test scenes of the GrandTour dataset. Each validation scene was recorded at the site of the test scene in the same column. Ours (w/o backend) is the frontend alone, and Sub. ID is the Codabench submission. **Bold** is the best and underline the second best.

Split	Validation							Test							
Scene	ARC-3	ARC-6	CON-3	EIG-2	SNOW-3	SPX-1	Avg.	ARC-2	ARC-7	CON-4	EIG-1	SNOW-2	SPX-2	Avg.	Sub. ID
FAST-LIVO2 [3]	1.34	1.24	0.97	2.59	3.44	1.64	1.87	1.22	1.38	3.13	3.32	1.61	2.28	2.16	880291
Ours (w/o backend)	<u>1.24</u>	0.89	1.00	2.10	1.43	<u>1.42</u>	1.35	0.86	1.19	2.30	2.21	1.41	1.99	<u>1.66</u>	933011
Ours	1.20	<u>0.91</u>	0.80	1.70	1.35	1.25	1.20	0.77	1.18	1.68	1.54	<u>1.42</u>	1.81	1.40	935484

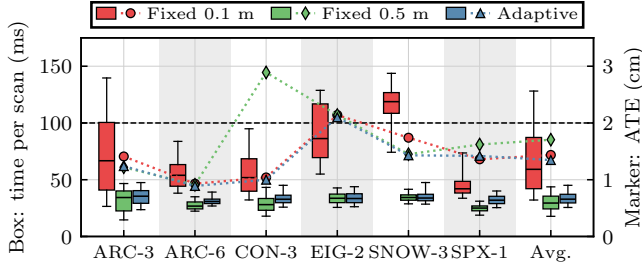


Fig. 2. Frontend time per scan (boxes, left axis) and ATE (markers, right axis) on the validation scenes with two fixed downsampling resolutions and our adaptive one. Boxes span the interquartile range and the lines the 5th to 95th percentiles. The dashed line is the scan period.

B. Backend

Plane bundle adjustment. The frontend registers each scan to the map near it, so the error between poses far apart in time accumulates. The backend corrects this drift with a global plane bundle adjustment [6] over submaps. Every 25 scans are merged into a submap at their frontend poses and downsampled to 0.2 m. Whenever four new submaps arrive, the backend runs the bundle adjustment. The adjustment divides the points of all submaps into 1 m voxels and keeps the voxels whose points form a plane. It then optimizes the submap poses so that the points in each voxel fit their plane.

Incremental pose graph. The backend output is a pose graph solved with iSAM2 [7]. Frontend poses give odometry factors within a sliding window of 10 scans. Each bundle adjustment gives an inter-submap factor, the relative pose between every two submaps. A gravity factor from the mean accelerometer reading is added to the scans where the robot stands still. The graph is updated incrementally as these factors arrive, and the latest estimate is output as the pose at 10 Hz.

III. RESULTS

Setup. We use eight cores of an Intel Core i9-13900 with 16 GB of RAM. All scenes run with one configuration except the IMU time offset, estimated per scene from the gyroscope and the trajectory rotation. We report the ATE RMSE in cm with the rule of the benchmark scorer.

Accuracy. Table I lists the ATE on the six test scenes and on their paired validation scenes from the same sites. The backend lowers the average ATE from 1.66 cm to 1.40 cm on the test scenes. The improvement is largest on EIG and CON, horizontally on EIG and in height on CON.

Adaptive downsampling. We compare the adaptive resolution with fixed ones in Fig. 2. The adaptive resolution matches the accuracy of the best fixed value, while a fixed resolution of 0.5 m triples the error on CON-3. The adaptive

TABLE II. Runtime on the test scenes, bags played at real time on 8 cores and 16 GB. A round is one bundle adjustment, and 100 ms is the scan period.

Scene	ARC-2	ARC-7	CON-4	EIG-1	SNOW-2	SPX-2
Frontend, mean (ms)	30.1	31.8	42.3	31.6	33.5	27.8
Scans over 100 ms	0	0	10	0	0	0
Backend, longest round (s)	1.4	1.4	19.4	4.4	3.1	0.7
Peak memory (GB)	9.0	7.5	16.7	11.0	6.2	7.9

time per scan stays near 35 ms within and across the scenes, whereas a fixed resolution of 0.1 m varies with the scene and exceeds the scan period on most scans of SNOW-3 and on many of EIG-2.

Runtime. We assign four threads to the frontend and the other four to the backend. The frontend takes 28 to 42 ms per scan on the twelve scenes of Table I. Table II lists the runtime on the test scenes. On five of the six test scenes, each bundle adjustment finishes within the 10 s before the next one. The later rounds on CON-4 take up to 19 s, so the correction arrives late but the output is not delayed.

TABLE III. Information requested by the organizers.

Item	Ours
Affiliation	Academic (Urban Robotics Lab, KAIST)
Compute	8 cores of an Intel Core i9-13900, 16 GB RAM
Modalities	Two LiDARs, one camera, one IMU
Topics	/boxi/hesai/points, /boxi/livox/points, /boxi/alphasense/front_center/image_raw/compressed (+camera_info), /boxi/stim320/imu, /tf_static
Code	Open source, github.com/gunheeShin/se3-livom-comfort

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